

1 The actual full good-copy density

The physical factors are Q, Y, V , with Y an independent middle factor; the auxiliary factors are C, R . The reduction below retains all five factors on the good copies. In particular, every good Y coordinate is retained until the physical symmetric projection is applied.

Definition 1 (Five-factor copy coordinates). *Let Q, Y, V, C, R be coordinate sets and let $m \geq 0$. A configuration x assigns one coordinate in each of these five sets to each copy j . The equivalence e_m groups these coordinates as*

$$e_m(x) = (a, (c, r)), \quad (1)$$

$$a_j = (x_{j,Q}, (x_{j,Y}, x_{j,V})), \quad (2)$$

$$c_j = x_{j,C}, \quad r_j = x_{j,R}. \quad (3)$$

Thus its range is

$$(Q \times (Y \times V))^m \times (C^m \times R^m). \quad (4)$$

The inverse places the five coordinates back in the same copy. This includes $m = 0$.

Definition 2 (Actual full good-copy density). *Let all five coordinate sets be finite. Put $A = Q \times (Y \times V)$, let $\Omega \in \mathbb{C}^A$, and let*

$$u \in \mathbb{C}^{A^k} \otimes \mathbb{C}^{C^k} \otimes \mathbb{C}^{R^k} \quad (5)$$

be arbitrary. For $B \subseteq \{0, \dots, k-1\}$, put $G = B^c$, $m = |G|$ and $b = |B|$. Form the actual component $w = (R_B \otimes I)u$. Use the same chosen enumeration of G in all five factors and the chosen enumeration of B in each auxiliary factor.

For a retained configuration x , write $e_m(x) = (a, (c, r))$. A discarded configuration consists of a physical assignment $d \in A^B$ and auxiliary assignments $c' \in C^b$, $r' \in R^b$. Reconstruct the original physical register by placing a on G and d on B . Reconstruct the two auxiliary registers by placing (c, r) on G and (c', r') on B . Let $f_B(x, (d, c', r'))$ be the value of w at these reconstructed coordinates. Define

$$\rho_G(x, x') = \sum_{d, c', r'} f_B(x, (d, c', r')) \overline{f_B(x', (d, c', r'))}. \quad (6)$$

This is the actual unnormalized rank-one partial trace. Only the bad physical and auxiliary copies are discarded. No condition on Ω or u is imposed in the definition.

Theorem 1 (Physical symmetric support of the full good-copy density). *In the preceding definition suppose $\|\Omega\| = 1$. Let P_{QYV} be the average of the simultaneous copy-permutation operators on the physical factors Q, Y, V , acting trivially on C, R . Then*

$$P_{QYV} \rho_G = \rho_G. \quad (7)$$

The original vector u need not be symmetric, unit or nonzero. Zero components, zero copies and empty good sets are included.

Proof. First trace only the bad physical copies of $|w\rangle\langle w|$, obtaining M on the good physical copies and the whole auxiliary register. The physical-support identity gives

$$(P_m \otimes I)M = M, \tag{8}$$

where P_m symmetrizes the m copies of A . Regroup each auxiliary register into its good and bad coordinates and trace the bad coordinates. Interchanging the finite sums shows that physical left multiplication passes through this partial trace. Hence the resulting grouped density N satisfies

$$(P_m \otimes I_{C^m \times R^m})N = N. \tag{9}$$

The same finite-sum interchange identifies N with ρ_G expressed in the coordinates e_m .

For each copy permutation, e_m transforms its physical-only action into the copy permutation of A^m tensored with the auxiliary identity. This follows directly from the coordinate entries. Averaging these identities transforms P_{QYV} into $P_m \otimes I$. Returning to the five-factor coordinates proves the assertion. $\square$

This establishes the actual physical symmetric-support step used in [1, Section 7], retaining the independent middle physical factor. The compatible physical spectral projection and the further reduction that traces the good Y coordinates are separate assertions.

References

- [1] OpenAI, *A two-dimensional area law from a global spectral gap*, September 24, 2026.